

FRANK PORTER
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M1 MACRO

Group B

Session 13

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

Dynamic Labor Demand Model

Price of the good: 1

$$\bullet F(L_t) = \alpha_0 L_t - \frac{\alpha_1}{2} L_t^2$$

$$F'(L) = \alpha_0 - \alpha_1 L_t \quad : (\alpha_0 \text{ large enough s.t. } F' > 0)$$

$$F'' = -\alpha_1 < 0$$

• $W_t = \text{wage (= labor cost)}$

• Adjust costs to labor: $\dot{C}(\Delta L_t) = \frac{1}{2} (\Delta L_t)^2$

$$1+n = \text{interest rate} = \frac{1}{\beta} \quad \Delta L_t = L_t - L_{t-1}$$

From maximizer

$$V_t = \sum_{i=0}^{\infty} \underbrace{\left(\frac{1}{1+r} \right)^i}_{\beta^i} \left[\alpha_0 \underline{L_{t+i}} - \frac{\alpha_1}{2} \underline{L_{t+i}^2} - W_{t+i} \underline{L_{t+i}} - \frac{b}{2} \left(\underline{L_{t+i}} - \underline{L_{t+i-1}} \right)^2 \right]$$

$$\left[\begin{array}{l} \text{Max } V_t \\ \{L_{t+i}\}_{i=0}^{\infty} \end{array} \right]$$

For wrt L_{t+i}^* :

$$\cancel{\beta^i} (\alpha_0 - \alpha_1 L_{t+i} - W_{t+i} - b(L_{t+i} - L_{t+i-1})) + \cancel{\beta^{i+1}} \cdot b(L_{t+i+1} - L_{t+i}) = 0$$

Write the FOC for $i=0$ (for simplicity)

(A) $\alpha_0 - \alpha, L_t - b(L_t - L_{t-1}) + \beta b(L_{t+1} - L_t) = W_t$

Sketch state of the problem.

Assume that $W_t = \bar{W} \quad \forall t$.

What is the $\bar{L}$ s.t. $L_t = \bar{L} \quad \forall t$ is
a solution of (A)?

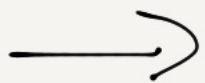
• If $b=0$ No adjust. costs

$$\text{FOC (A)} : \alpha_0 - \alpha_1 L_t = W_t$$

$$\Leftrightarrow \boxed{L_t = \frac{\alpha_0 - W_t}{\alpha_1}}$$

($\alpha_0 - W_t > 0$
because α_0
is "large")

↳ STATE DECISION



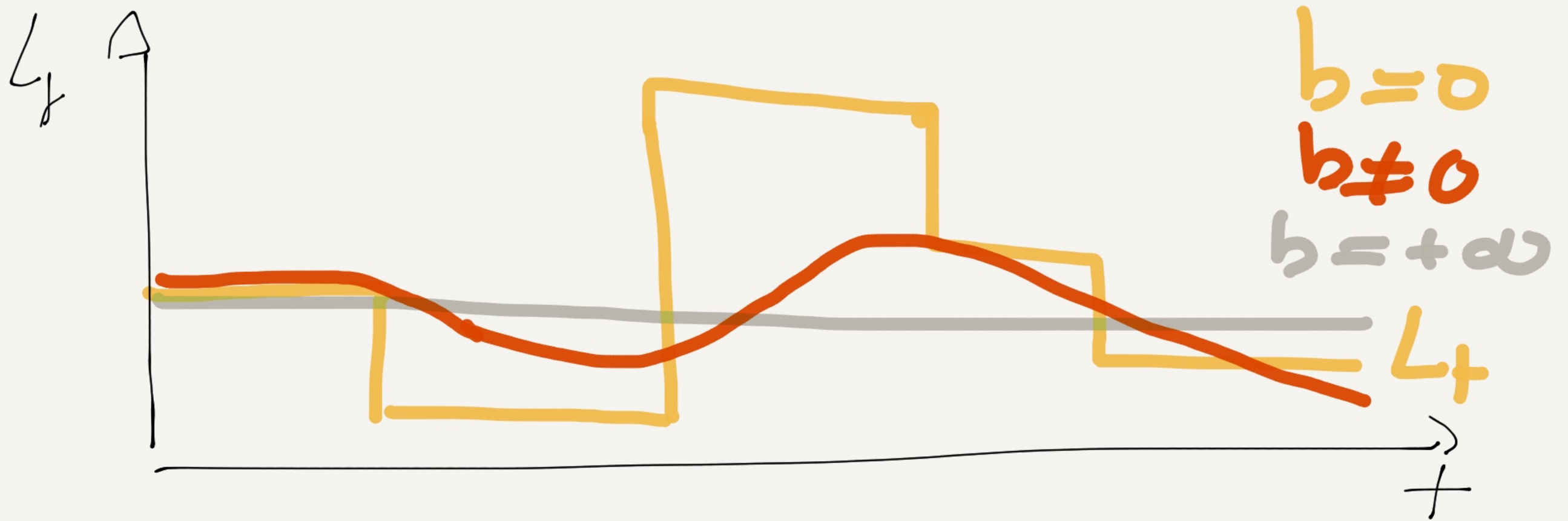
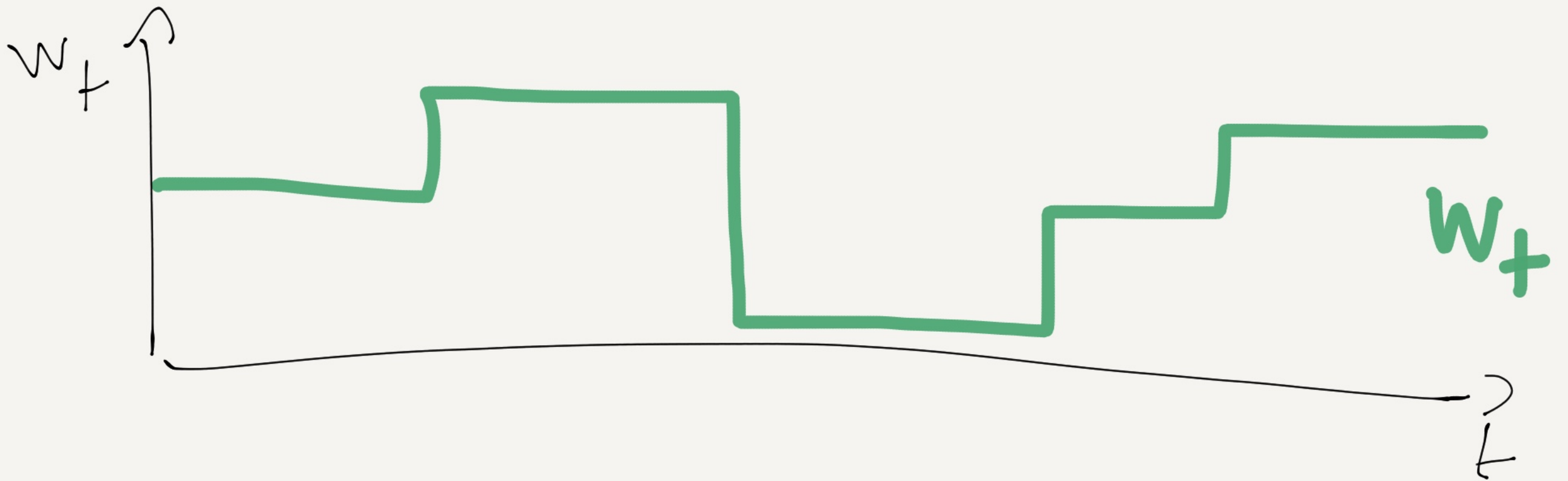
• back to $b \neq 0$

Steady State: $L_t = \bar{L} \quad \forall t$ (and $W_t = \bar{W}$)

$$(A) \quad \alpha_0 - \alpha_1 \bar{L} - \bar{W} - b(\bar{L} - \bar{L}) + b(\bar{L} - \bar{L}) = 0$$

$$\bar{L} = \frac{\alpha_0 - \bar{W}}{\alpha_1}$$

Steady State



Solve for the optimal path $\{L_t\}$

Assume $W_t = \overline{W} \quad \forall t$

To solve

$$(A) \quad \alpha_0 - \alpha_1 L_t - b(L_t - L_{t-1}) + \beta b(L_{t-1} - L_t) = \overline{W}$$

change of variable $\therefore \overline{L} = \frac{\alpha_0 - \overline{W}}{\alpha_1}$

$\hat{L}_t = \text{distance from the SS} = L_t - \overline{L}$

$$\begin{aligned}
 \textcircled{A} \quad & \alpha_0 - \alpha_1 (\overbrace{l_t - \bar{l} + \bar{l}}^{\hat{l}_t}) - b (\overbrace{l_t - \bar{l} + \bar{l}}^{\hat{l}_t} - \overbrace{l_{t-1} - \bar{l} + \bar{l}}^{\hat{l}_{t-1}}) \\
 & + \beta b (\overbrace{l_{t+1} - \bar{l} + \bar{l}}^{\hat{l}_{t+1}} - \overbrace{l_t - \bar{l} + \bar{l}}^{\hat{l}_t}) = \bar{w}
 \end{aligned}$$

$$\Leftrightarrow \alpha_0 - \alpha_1 \hat{l}_t - \alpha_1 \bar{l} - b (\hat{l}_t - \hat{l}_{t-1}) + b (\hat{l}_{t+1} - \hat{l}_t) = \bar{w}$$

$$\alpha_0 - \alpha_1 \bar{l} = \alpha_0 - \alpha_1 \left(\frac{\alpha_0 - \bar{w}}{\alpha_1} \right) = \bar{w}$$

↳ I can equivalently solve

$$-\alpha_1 \hat{l}_t - b (\hat{l}_t - \hat{l}_{t-1}) + \beta b (\hat{l}_{t+1} - \hat{l}_t) = 0 \quad \textcircled{AA}$$

$$(AA) \Leftrightarrow \boxed{\beta b \hat{L}_{t+1} - (\alpha_1 + b + \beta b) \hat{L}_t + b \hat{L}_{t-1} = 0}$$

Math property : $a x_{t+1} + b x_t + c x_{t-1} = 0 \quad (A)$

The solution of (A) is

$$x_t = c_1 \lambda_1^t + c_2 \lambda_2^t$$

where (λ_1, λ_2) are the roots (real or complex) of
the characteristic polynomial $P(\lambda) = a\lambda^2 + b\lambda + c$

c_1 and c_2 are determined by two boundary conditions

[for ex: x_0, x_1 given]

$$\begin{cases} x_0 = c_1 \times t_1^0 + c_2 \times t_2^0 = c_1 + c_2 \\ x_1 = c_1 t_1 + c_2 t_2 \end{cases}$$

→ 2 eq. in (c_1, c_2)

$$\beta b \Gamma_{t+1} - (d_1 + b + \beta b) \Gamma_t + b \Gamma_{t-1} = 0$$

Characteristic polynomial : $P(\lambda) = \beta b \lambda^2 - (d_1 + b + \beta b) \lambda + b = 0$

λ_1, λ_2 such that $P(\lambda_1) = P(\lambda_2) = 0$.

Math property : $ax^2 + bx + c = 0$, Root : x_1, x_2

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x_1 + x_2 = -\frac{b}{a}$$

$$x_1 \times x_2 = \frac{4ac}{4a^2} = \frac{c}{a}$$

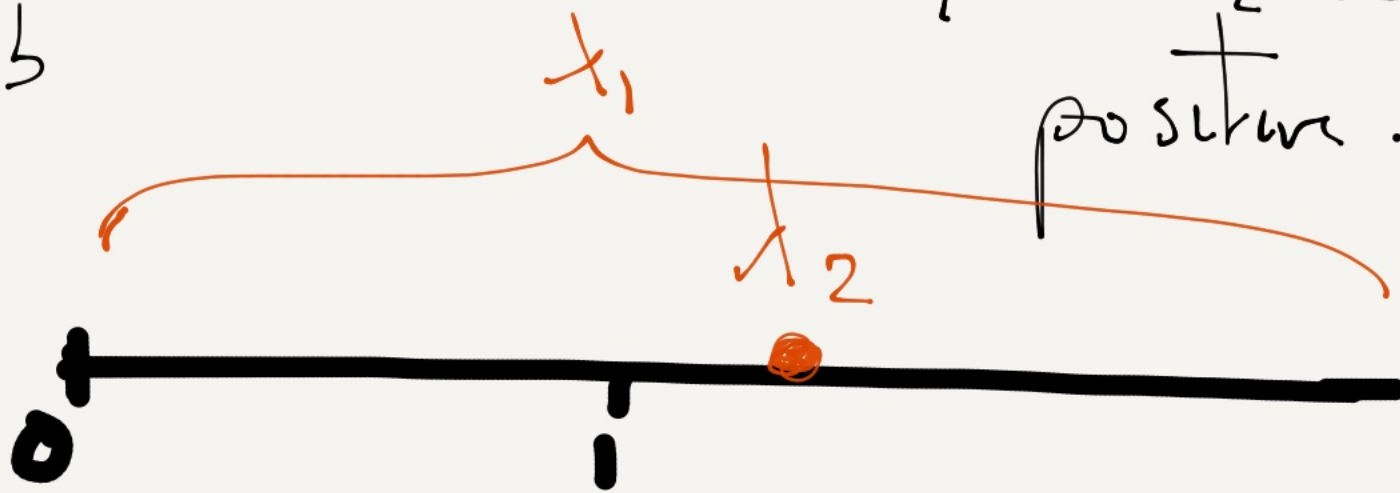
Here

$$\lambda_1 \times \lambda_2 = \frac{1}{\beta} > 1 > 0 \quad \left(\beta = \frac{1}{1+r} < 1 \right)$$

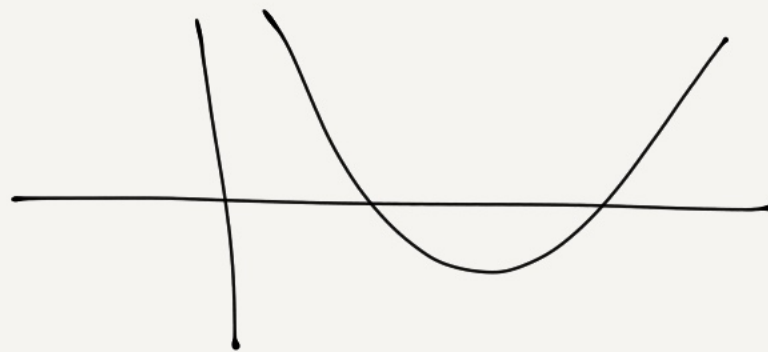
$(r > 0)$
 $\hookrightarrow$ The 2 roots have the same sign ($\lambda_1 \lambda_2 > 0$)
• at least one root is bigger than 1 in modulus

$$\bullet \lambda_1 + \lambda_2 = \frac{a_1 + b + \beta b}{\beta b} > 0 \Rightarrow \lambda_1 \text{ and } \lambda_2 \text{ are positive.}$$

$\hookrightarrow$ I have proved:



$P(\lambda)$ is



because

$$\beta b > 0$$

$$P(1) = \beta b - \alpha_1 - \beta b - b + b = -\alpha_1 < 0$$

